



The Mariana Trench

The deepest known point on the seabed of planet Earth, created by the collision of two oceanic plates

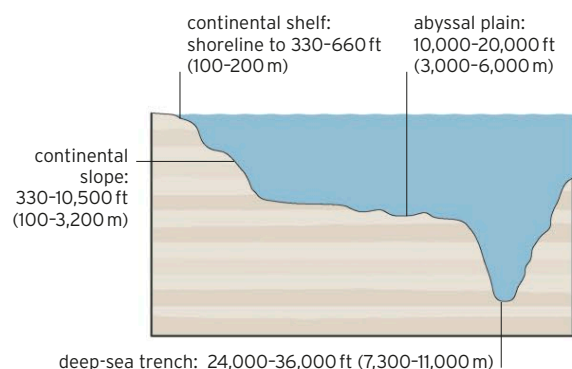
On the floor of the western Pacific Ocean, around 200 miles (322 km) southwest of the island of Guam, the Mariana Trench plunges to a staggering 36,201 ft (11,034 m) at a section called the Challenger Deep—the deepest point yet discovered in any of Earth's oceans. If the world's highest mountain, Mount Everest, were placed inside the trench, it would still be covered by 7,162 ft (2,183 m) of water.

When plates collide

The Mariana Trench marks the location of a subduction zone, the point at which one oceanic tectonic plate sinks under another. Here, the huge Pacific Plate has plunged so dramatically beneath the much smaller Philippine Plate and Mariana Microplate that it has created the steep-sided trench, which, at over 1,580 miles (2,540 km) long, is more than five times longer than the Grand Canyon yet averages just 43 miles (69 km) wide.

PARTS OF THE OCEAN FLOOR

Just like landmasses, seabeds contain plains, ridges, mountains, and canyons. From the shoreline, the continental shelf inclines gradually until it drops away sharply into the deep, open waters of the continental slope. Where there is no mid-ocean ridge system, this levels out into the abyssal plain before plunging into deep-sea trenches such as the Challenger Deep.



▷ WHITE SMOKE

Undersea volcanic activity creates hydrothermal vents along the Mariana volcanic arc, such as this white smoker—the so-called Champagne Vent—which releases bubbles of liquid carbon dioxide.



Remnant arc

The remnant arc once contained active volcanoes, but it has since been pushed away from eruptive areas and is now inactive.

Philippine Plate

Although the Philippine Plate is overriding the Pacific Plate, the upwelling mantle results in spreading, and the plate moves away from the subduction zone.

► SUBDUCTING PLATE

These cross-sections of the Mariana Trench subduction zone show many of the geological processes involved. As the two oceanic plates converge, one pushes beneath the other, creating the trench itself but also setting in motion the processes that produce chains of seamounts and islands, and new ocean floor.

Active back-arc basin

Back-arc basins are created behind island arcs as a result of rifting and spreading of the sea floor due to the huge forces created by the presence of a deep ocean trench.

Remnant arc

When volcanic activity at an undersea ridge ceases, the ridge is known as a remnant arc.

seamount, an inactive submarine volcano

oceanic crust

lithospheric mantle

asthenosphere

Mariana microplate movement

The Mariana island arc rests on a tectonic microplate called the Mariana Plate. Located between sections of the Philippine and Pacific Plates, it is diverging from the former, while the latter is subducting beneath it at the Pacific Plate's southeastern end.

The Challenger Deep

Despite total darkness, extremely cold water, and crushing pressure, the Challenger Deep is a habitat that supports a host of living creatures.

Subducting plate

As the Pacific Plate slides beneath the overlying plate at a rate of about $1\frac{1}{4}$ in (3 cm) per year, the plates rub against each other, generating tremors that can be felt across a wide area.